Bayesian Estimators

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Data

Let $X_1, \dots, X_n \overset{iid}{\sim} F(\theta)$ where $F(\cdot)$ is a known distribution function and θ is a vector of parameters. Let $X = (X_1, \dots, X_n)^{\mathrm{T}}$, be the sample collected.

Frequentist-based Estimation

Bothe the **maximum likelihood estimator** and **method of moments estimator** are known as frequentist-based estimators since the parameters we are esimating are considered **fixed**.

Bayesian Estimation

The idea of Bayesian estimation is that our parameters θ are random variables instead. Therefore; they follow the probability theory that we have learned so far.

Using the prior distribution, we would like to find the distribution of θ when we incorporate the data X into our distribution. We are looking to find the **posterior distribution**.

$$\theta | X \sim F(\gamma)$$

Prior Distribution

The prior distribution is the distribution for the random variable θ:

$$\pi(\theta)$$

Likelihood Function

$$L(\theta|x) = f(x|\theta) = \prod_{i=1}^{n} f(x_i|\theta)$$

Likelihood X Prior

$$f(\theta, x) = L(\theta|x) \times \pi(\theta)$$

What is $f(\theta|x)$?

$f(\theta, x)$ want $f(\theta | x)$

$$\frac{f(\theta, x)}{f(x)}$$

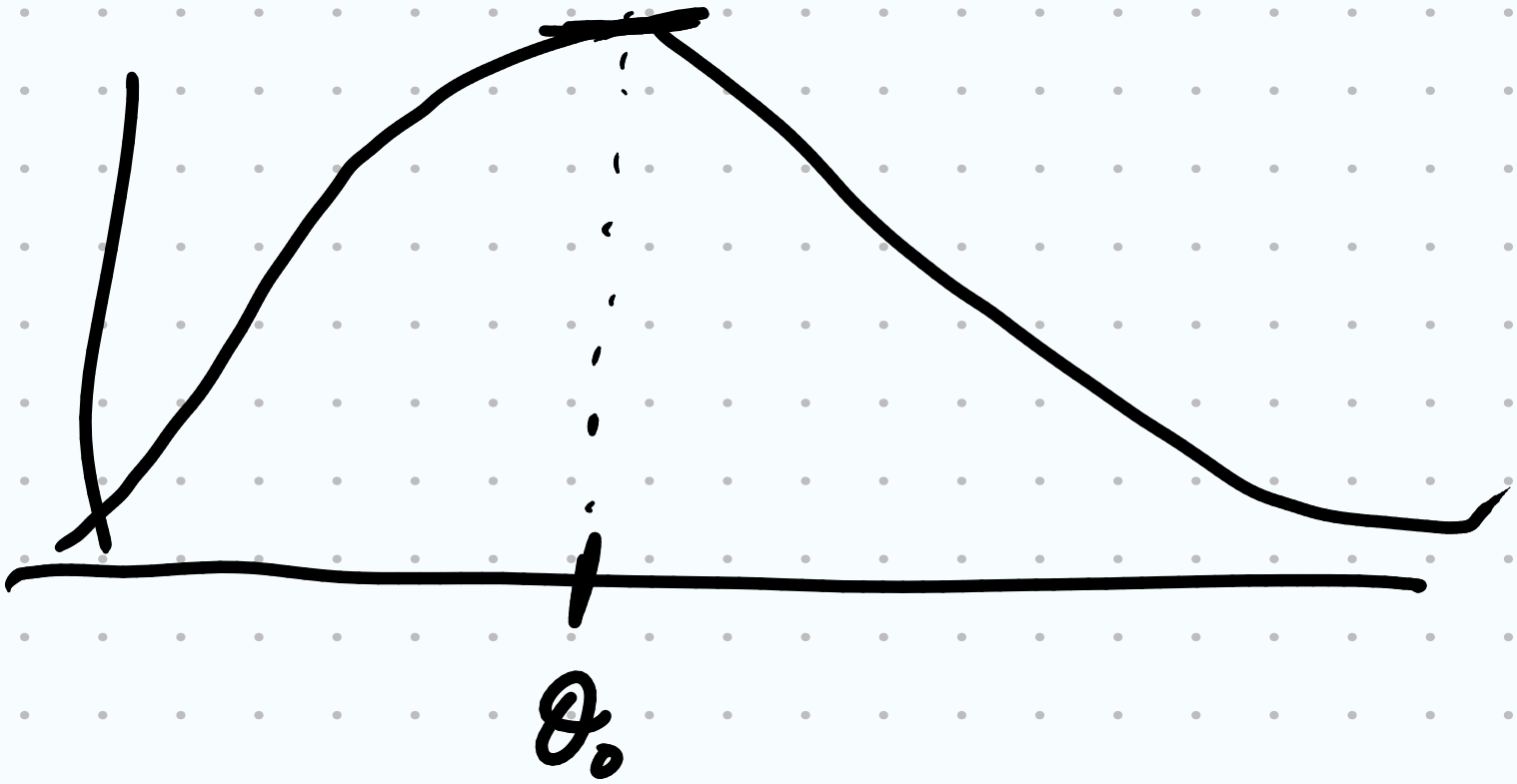
$$f(x) = \int_{\theta} f(\theta, x) \, d\theta$$

Posterior Distribution

The posterior distribution is the distribution of θ when we condition it on the data X

$$\pi(\theta|x) = \frac{L(\theta|X)\pi(\theta)}{\int L(\theta|X)\pi(\theta)d\theta}$$

$\pi(\theta|x)$ $E(\theta|x)$



Estimators

- Bayesian Estimator
- *maximum a posteriori* (MAP)

Bayesian Estimator

$$\widehat{\theta} = E_{\theta|X}(\theta) = \int \theta \pi(\theta|x) d\theta$$

MAP

$$\tilde{\theta} = \mathrm{argmax}_{\theta} \pi(\theta|x)$$

Most people implement a Markov Chain Monte Carlo Method to construct the posterior distribution.

They use sample means to approximate $E(\theta|x)$

Conjugate Prior

A conjugate prior distribution is any distribution that result in the posterior distribution having a similar form as the conjugate distribution.

Hyper Parameters

Hyperparameters are the parameters involved in the prior distribution.

Bernoulli Distribution

Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Bin}(1, p)$

p

$$L(p|x) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i}$$

$$L(p|x) = p^{n\bar{x}}(1-p)^{n-n\bar{x}}$$

$$Y \sim Bek(\alpha, \beta)$$

$$f(y) = \frac{1}{B(\alpha,\beta)} y^{\alpha-1}(1-y)^{\beta-1}$$

$$p \sim Bek(\alpha, \beta)$$

$$p^{n\bar{x}}(1-p)^{n-n\bar{x}} \frac{1}{B(\alpha,\beta)} p^{\alpha-1}(1-p)^{\beta-1}$$

$$\frac{1}{B(\alpha,\beta)} p^{n\bar{x}+\alpha-1} (1-p)^{n-n\bar{x}+\beta-1} = f(p,x)$$

$$f(x) = \int f(p,x)\, dp$$

$$f(p|x) = f(p,x) / f(x)$$

$$f(p|x) \propto p^{n\bar{x}+\alpha-1} (1-p)^{n-n\bar{x}+\beta-1}$$

$$p|x \sim \text{Beta}(n\bar{x}+\alpha, n-n\bar{x}+\beta)$$

$$\int_0^1 k\, p^{n\bar{x}+\alpha-1}(1-p)^{n-n\bar{x}+\beta-1}\, dp = 1$$

solve for k

$$E(p|x) = \frac{n\bar{x}+\alpha}{n\bar{x}+\alpha+n-n\bar{x}+\beta} = \frac{n\bar{x}+\alpha}{\alpha+\beta+n}$$

Poisson Distribution

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Find the likelihood.

Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Pois}(\lambda)$

$$L(\lambda \mid x) = \prod \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$= e^{-n\lambda} \lambda^{n\bar{x}} \prod \frac{1}{x_i!}$$

$$f(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-y\beta}$$

β rate instead of scale

scale: $\frac{1}{\beta}$

$$\lambda \sim \text{Gamma}(\alpha, \beta)$$

$$e^{-n\lambda} \lambda^{n\bar{x}} \prod \frac{1}{x_i!} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\lambda\beta} = f(\lambda, x)$$

$$\frac{\beta^\alpha}{\Gamma(\alpha)} \prod \frac{1}{x_i!} e^{-n\lambda - \lambda\beta} \lambda^{n\bar{x}+\alpha-1}$$

$$f(\lambda|x) \propto e^{-\lambda(n+\beta)} \lambda^{n\bar{x}+\alpha-1}$$

$$\lambda|x \sim \text{Gamma}(n\bar{x}+\alpha, n+\beta)$$

$$E(\lambda|x) = \frac{n\bar{x}+\alpha}{n+\beta}$$

Nomal Distribution

σ^2 is known

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$

$$L(\mu | x) = \prod \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

$$= (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2}\sum(x_i-\mu)^2}$$

$$\sum(x_i-\mu)^2$$

$$\sum x_i^2 - 2x_i\mu + \mu^2$$

$$\sum x_i^2 - 2\mu n\bar{x} + n\mu^2$$

$$n\mu^2 - 2n\mu\bar{x} + \sum x_i^2$$

$$(2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2}\left[n\mu^2 - 2n\mu\bar{x} + \sum x_i^2\right]}$$

$$(2\pi\sigma^2)^{-n/2} e^{-\frac{\Sigma x_i^2}{2\sigma^2}} e^{-\frac{1}{2\sigma^2}\left[n\mu^2 - 2n\mu\bar{x}\right]}$$

$$(2\pi\sigma^2)^{-n/2} e^{-\frac{\Sigma x_i^2}{2\sigma^2}} e^{-\frac{n}{2\sigma^2}\left[\mu^2 - 2\mu\bar{x} - \bar{x}^2 + \bar{x}^2\right]}$$

$$\mu^2 - 2\mu\bar{x} - \bar{x}^2 + \bar{x}^2$$

$$(2\pi\sigma^2)^{-n/2} e^{-\frac{\Sigma x_i^2}{2\sigma^2}} e^{+\frac{n\bar{x}^2}{2\sigma^2}} e^{-\frac{n}{2\sigma^2}(\mu-\bar{x})^2}$$

$$e^{-\frac{(\mu-\bar{x})^2}{2\sigma^2/n}}$$

$$\mu \sim N(\theta, \tau^2)$$